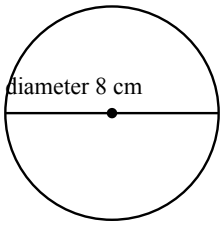
Circles and first previews

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. A circle has diameter 8 cm. Find its circumference in terms of π and using 3.14.



P2. A circle has radius 3 cm. Find its diameter and circumference in terms of π.

Answer: _____

P3. Find the area of a circle with radius 4 cm. Give an exact answer and an approximation using 3.14.

Answer: _____

Circles and first previews

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. A circle has diameter 6 cm. Find its area in terms of π and using 3.14.

Answer: _____

P5. Circle A has diameter 4 cm and circumference 4π cm. Circle B has diameter 10 cm and circumference 10π cm. Compare C ÷ d.

Answer: _____

P6. A collection grows from 80 cards to 100. Find the percent increase. Then find the percent decrease from 100 back to 80.

Answer: _____

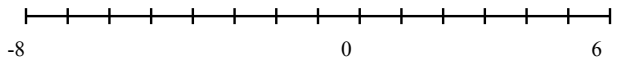
Circles and first previews

Follow the slide labels. P = problem; F = your turn; D = discussion.

P7. Start with 80 tokens. Increase by 25%, then decrease the result by 25%. Find the final amount. Explain why it is not 80.

Answer: _____

P8. Use steps on a number line for −2 + 5, −3 + (−4), and 2 − 6.



F. A circle has diameter 16 cm. Complete: radius = __; area = π × __² = __π square cm.

Answer: _____

Circles and first previews

Follow the slide labels. P = problem; F = your turn; D = discussion.

D. Maya finds area from a diameter of 14 cm by writing $\pi \times 14^2 = 196\pi$. Explain and repair it.

Answer: _____

S. For circle area, I first identify __. A percent change compares the change with __.

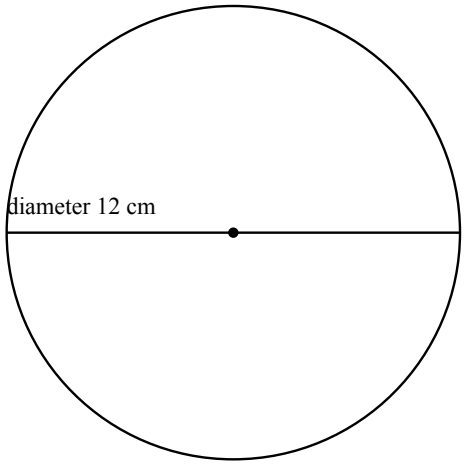
Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. A circle has diameter 12 cm. Find its area in terms of π and using 3.14. Show the radius you used.

Answer: _____



Final Mixed Review

15 minutes. No calculator. Show your work.

1. Compare $\frac{5}{8}$ and $\frac{3}{5}$.

Answer: _____

2. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

Answer: _____

3. A machine makes 24 labels in 6 minutes. Find labels per minute and minutes per label.

Answer: _____

4. Share $\frac{5}{6}$ liter equally among 4 cups. How much goes in each cup?

Answer: _____

5. Find $9 \div 40$.

Answer: _____

Final Mixed Review

6. Expand $2(x + 7)$.

Answer: _____

7. A prism has right-triangle ends with sides 3, 4, 5 cm and length 4 cm. Find surface area.

Answer: _____

8. Find the opposite of -6.

Answer: _____

9. Find Q1, Q3, and IQR for 0, 2, 3, 5, 7, 9, 10, 12. Exclude the median if the count is odd.

Answer: _____

10. A circle has diameter 3 cm. Find circumference exactly and using 3.14.

Answer: _____

Scale, proportions, circles, and review

The big idea

A fixed multiplier connects scaled lengths and proportional quantities. Circle formulas use radius or diameter in different ways. The final weekdays revisit the entire program.

Words to know

scale factor: The multiplier from an original length to a copy length. Example: A factor of 2 doubles every length.

scaled copy: A shape with matching angles and one multiplier for every length. Example: A 2 by 3 rectangle becomes 4 by 6.

constant of proportionality: The same y per one x in a proportional relationship. Example: For $y = 4x$, $k = 4$.

proportional: Related by a fixed multiplier with zero matched to zero. Example: At $3 each, cost = 3 × item count.

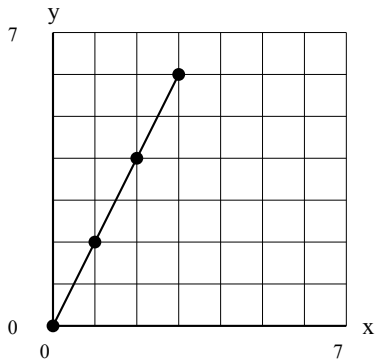
radius: Distance from the center to the circle. Example: A radius of 3 cm gives diameter 6 cm.

diameter: Distance across the circle through its center. Example: Diameter = 2 × radius.

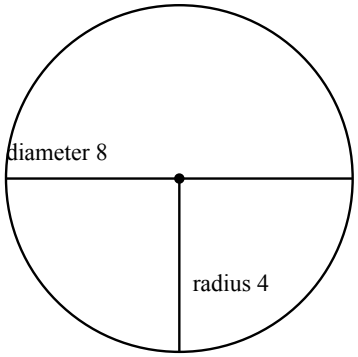
circumference: The distance around a circle. Example: $C = \pi d$

π: The ratio of a circle's circumference to its diameter. Example: π is about 3.14, not exactly 3.14.

How it works



The straight line y = 2x passes through the origin. A scaled copy multiplies every length by the same factor. Proportional relationships have the form y = kx.



A diameter contains two radii. Circumference measures the boundary. Circle area uses the square of the radius. π is exact; 3.14 is an approximation.

Worked examples

1. A copy of a 3 cm by 5 cm rectangle uses scale factor 2.

The new lengths are 6 cm and 10 cm.

The original area is 15 square cm; the copy area is 60 square cm.

The area factor is $60 \div 15 = 4 = 2^2$.

Answer: 6 by 10 cm; area 60 square cm

2. Find area when a circle's diameter is 10 m.

The radius is half the diameter: 5 m.

Square the radius: $5^2 = 25$.

Area = 25π square m, or about 78.5 square m using 3.14.

Answer: 25π square m; about 78.5 square m

Common mistakes

Adds the same amount to every length. Why: Confuses additive change with scaling. Fix: Compare each copy-to-original ratio.

Uses the length factor for area. Why: Counts one changing direction. Fix: Build an array of scaled unit squares.

Calls any straight line proportional. Why: Misses the origin requirement. Fix: Find the output when the input is zero.

Optional preview

Signed multiplication follows equal-step patterns: $(-3) \times 4 = -12$. A quotient asks for the missing factor. Since $(-3) \times 4 = -12$, $(-12) \div (-3) = 4$.

Try it

1. A 3 by 4 cm rectangle is copied with scale factor 2. Find the new side lengths.

2. A 5 by 6 cm rectangle is copied with scale factor 3. Find the new side lengths.

3. A circle has diameter 3 cm. Find circumference exactly and using 3.14.

4. A circle has diameter 5 cm. Find circumference exactly and using 3.14.

5. A circle has diameter 7 cm. Find circumference exactly and using 3.14.

Answers

1. 6 cm and 8 cm 2. 15 cm and 18 cm 3. 3π cm; about 9.42 cm 4. 5π cm; about 15.70 cm 5. 7π cm; about 21.98 cm

Where this connects

The bridge joins ratios, equations, and geometry. Future lessons can revisit these ideas with more complex percent and signed-number problems.